



Practice Set-01: Descriptive Statistics

STA201 – Theory of Probability and Statistics

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Dr. Lokesh Singh

Department of Mathematics,
Amity University Punjab, Mohali.

lsingh@pb.amity.edu

Instructions

Answer all questions by creating an appropriate table. *The final answers are provided at the end of the document for self-assessment.*

1. The diameters (in mm) of ten ball bearings produced by a machine are as follows:

12.5, 12.8, 12.3, 12.9, 12.4, 12.7, 12.6, 12.5, 12.8, 12.5.

Compute:

- (a) mean
- (b) median
- (c) mode
- (d) range
- (e) mean deviation about the mean
- (f) standard deviation
- (g) coefficient of variation

2. The number of defective transistors in a box of 50 is noted for 100 boxes. The frequency distribution is:

Defects (x) :	0	1	2	3	4
Boxes (f) :	14	32	28	18	8

Determine:

- (a) mean
- (b) median
- (c) mode
- (d) standard deviation

3. The life (in hours) of 80 electric bulbs is recorded as:

Life (hrs) :	0 – 200	200 – 400	400 – 600	600 – 800	800 – 1000	1000 – 1200
Frequency :	8	12	20	25	10	5

Using the step-deviation method, calculate the mean. Also find the median and mode.

4. The mean of six numbers is 15. If one number is excluded, the mean of the remaining five numbers becomes 13. Find the excluded number.
5. The mean of twenty observations is 25. Later it was discovered that an observation 30 was misread as 20. Find the correct mean.
6. The mean of forty items is 30. Two new items 35 and 45 are added. Find the new mean.
7. For a set of fifty numbers, the mean is 20 and standard deviation is 4. If every number is multiplied by 3, find the new mean and new standard deviation.

8. The mean of the following frequency distribution is 24. Find the missing frequency p .

Class :	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
Frequency :	5	10	p	8	2

9. Compute the mean deviation about the median for the ungrouped data:

6, 9, 12, 15, 18, 21, 24, 27.

10. Using the assumed mean method (take $A = 1250$), calculate the mean and standard deviation of the following tensile strengths (MPa):

1250, 1280, 1230, 1310, 1260.

11. The marks of 200 students are grouped as follows:

Marks :	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60
Students :	20	30	50	60	30	10

Using the step-deviation method, determine the mean, standard deviation, and coefficient of variation.

12. The daily wages of 100 workers are:

Wages (Rs) :	50 – 70	70 – 90	90 – 110	110 – 130	130 – 150
Workers :	15	25	30	20	10

Find:

- Pearson's coefficient of skewness (use the mode).
 - The moment coefficient of skewness γ_1 .
13. The second and fourth central moments of a dataset are $m_2 = 16$ and $m_4 = 1024$. Calculate the kurtosis β_2 and the excess kurtosis γ_2 . What does the result indicate about the distribution?
14. The number of power outages per year in a city over ten years is:

2, 1, 0, 3, 2, 4, 1, 2, 3, 2.

Compute the coefficient of kurtosis β_2 and interpret its value.

15. The mean and standard deviation of twenty observations are 50 and 5, respectively. Later it is found that an observation 45 was wrongly recorded as 55. Obtain the correct mean and correct standard deviation.
16. Each of the fifteen values in a dataset is increased by 20%. Determine the percentage change in

- (a) the mean,
- (b) the standard deviation,
- (c) the coefficient of variation.

17. In a distribution, the mean is 45, the mode is 42, and the standard deviation is 8. Find Pearson's coefficient of skewness. If every value is increased by 10, what will be the new skewness?
18. The median of the following frequency distribution is 32.5. Find the missing frequency.

Class :	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60
Frequency :	5	8	?	16	10	6

19. Compute the mean deviation about the mean for the grouped data:

Marks :	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
Students :	4	6	15	10	5

20. If a distribution is perfectly symmetric, what are the values of Pearson's coefficient of skewness and the moment coefficient of skewness γ_1 ?
21. For a dataset, the moment coefficient $\beta_1 = 0.49$. Determine γ_1 and interpret the direction of skewness (state the usual sign convention).
22. The range of a dataset is 40 and the smallest value is 10.
- (a) What is the largest value?
 - (b) Explain why the range is considered a poor measure of dispersion.
23. Find the coefficient of range for the data:

15, 22, 30, 18, 25, 40, 12.

24. A student's final grade is based on the following components:

- Homework: weight 20%, score 85
- Midterm: weight 30%, score 78
- Project: weight 10%, score 92
- Final Exam: weight 40%, score 80

Calculate the weighted mean score.

25. For a frequency distribution, the first three moments about the value 5 are $m'_1 = 2$, $m'_2 = 20$, and $m'_3 = 140$. Find:
- (a) the actual mean,
 - (b) the variance,
 - (c) the moment coefficient of skewness γ_1 .

Answers

1. (a) 12.60 mm (b) 12.55 mm (c) 12.5 mm (d) 0.6 mm (e) 0.16 mm (f) $\sigma \approx 0.1844$ mm (g) $\approx 1.46\%$
2. (a) 1.74 (b) 2 (c) 1 (d) $\sigma \approx 1.1456$
3. Mean = 580 hrs; Median = 600 hrs; Mode = 650 hrs
4. 25
5. 25.5
6. ≈ 30.476
7. New mean = 60; New SD = 12
8. $p = 55$
9. 6
10. Mean = 1266; SD ≈ 27.276
11. Mean = 29; SD ≈ 13.19 ; CV $\approx 45.48\%$
12. (a) $Sk_P \approx 0.0138$ (b) $\gamma_1 \approx 0.1148$
13. $\beta_2 = 4$; $\gamma_2 = 1$; leptokurtic
14. $\beta_2 = 2.5$; platykurtic (flatter than normal)
15. Correct mean = 49.5; Correct SD ≈ 4.975
16. (a) +20% (b) +20% (c) 0% (no change)
17. $Sk_P = 0.375$; remains 0.375
18. Missing frequency = 11
19. 8.875
20. Both coefficients = 0
21. $\gamma_1 = 0.7$; usually positive γ_1 indicates positive skew
22. (a) Largest = 50 (b) Range uses only the two extreme values and ignores the distribution of all other data; it is highly sensitive to outliers.
23. $\frac{28}{52} = \frac{7}{13} \approx 0.5385$
24. 81.6
25. (a) Mean = 7 (b) Variance = 16 (c) $\gamma_1 = 0.5625$